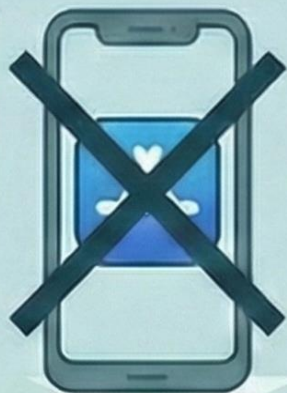


Safe Children Project (MVP)

Rapid Response Beach Safety System – Technology Overview

"No-App" Philosophy



Eliminates barriers to entry by removing the need for parents or the public to download a custom mobile application.

Universal Communication Channels



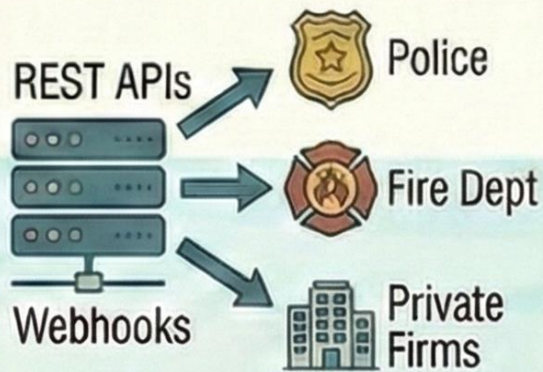
Primary reliance on SMS and standard Email for critical parent notifications to ensure delivery regardless of data coverage.

Web-Responsive Design



A "Mobile-First" web platform accessible via any standard browser on any device (smartphones, tablets, desktops).

API-Driven Architecture



Built for scalable integration with institutional partners (Police, Fire Dept, private beach management firms) via REST APIs and Webhooks.

Technologies and Hardware

Low-Cost Technologies and Infrastructure

Smart Wristbands

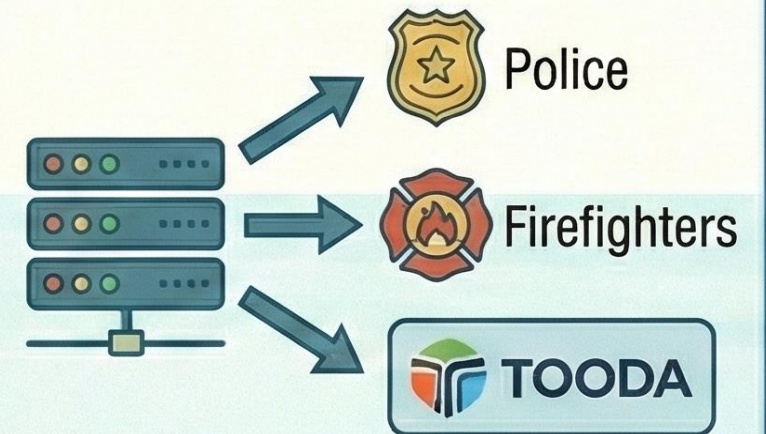


Disposable, waterproof, single-use paper wristbands containing unique pre-printed QR Codes.

Notification Channels



SMS & E-mail: Primary channels to ensure alert receipt even without stable internet.



Webhooks & API REST: Integration for institutional managers (Police, Firefighters, private partners like TOODA) to receive real-time data.

Decentralized Security Point Network

Concept



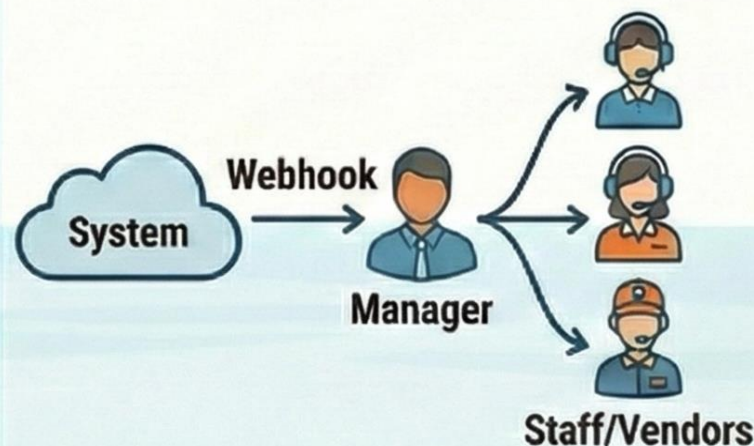
Security is managed by institutions (Managers) that oversee networks of physical points (kiosks, lifeguard stations).

Active Status Control



Security professionals can mark their status as "On Duty" (Active), ensuring only available personnel receive real-time alerts.

Hierarchy



The system alerts the Manager via Webhook, who then distributes the notification to their network of staff/vendors at the edge.

Registration & Check-in Flow

Bracelet Binding



Parents scan the security point QR and the bracelet QR to bind a child's profile to the system.

Required Documentation



Child's photo, name, age, and a primary SMS-capable phone number are mandatory for activation.

Location Tracking



When arriving at the beach, parents scan a security point QR to set the child as "active" at that specific location.

System Confirmation



The system sends an automated SMS/Email confirmation to the parent upon successful check-in or check-out.

Alert System (LOSS): Instant Mobilization in Case of Loss

Multiple Reporting Forms



By Guardian
(via logged-in
web app)

By Any Person
scanning the
public wristband
QR Code (with
prior informant
identification)

Intelligent Alert Distribution

Primary (Push)



Instant Webhook notifications to
Security Point Managers.

Secondary (Fallback)



Direct SMS to Firefighters, Security
Points, and registered contacts.

Visual Content

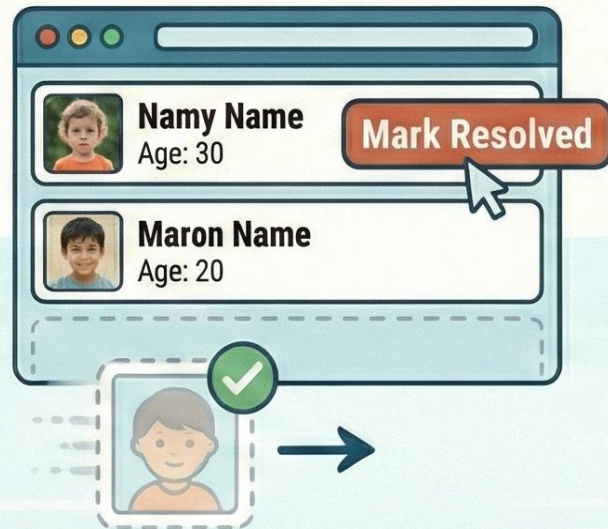


The alert includes a large
photo of the child, vital data,
and the last known location.

Security Forces Dashboard

Real-Time Tactical View

Active Cases Focus



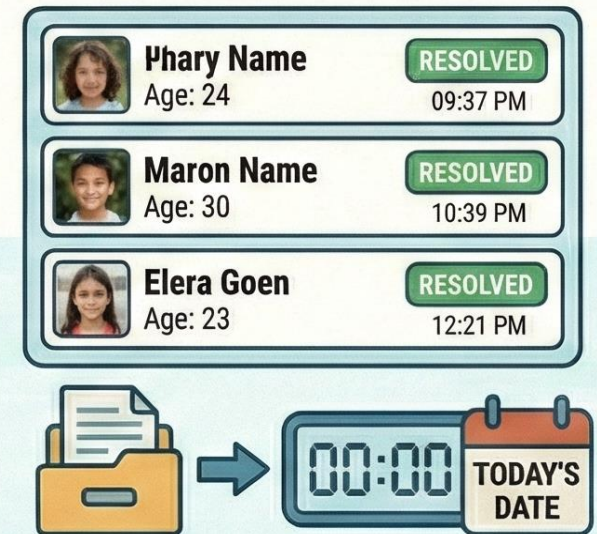
The dashboard exclusively displays currently missing children, removing cases once resolved to avoid visual clutter.

Live Monitoring



Automatic update every 30 seconds, with sound alerts for new cases and a timer indicating the time elapsed since disappearance.

Daily History



A separate section displays cases resolved on the current day, which are automatically archived at midnight.

Child Found and Closure Protocol

Secure Reunion and Case Closure

Finder Identification



Any citizen who finds the child must provide their name and phone number before viewing the parents' emergency contacts.

Immediate Parent Notification



Immediate SMS containing the finder's name, phone number, and exact location (e.g., 'Tower 28'), with a quick call button.

Alert Cancellation



After reunion confirmation, a cancellation notification is sent to all managers and security points, removing the alert from dashboards.